ALGEBRA 1 · UNIT 1

Section 1.5 Equations as Models

Equations can do more than give an answer. They can represent a situation and help us decide what a solution means.

> **Mathematical idea:** A useful equation model connects the operations, variables, units, and solution back to the real situation they came from.

Today we will practice defining variables, checking reasonableness, solving equations, interpreting solutions, and comparing equation models.

Learning Target

Students will write, solve, and interpret equations that model situations.

I can statements

- I can write equations to represent situations.
- I can solve equations and interpret solutions.
- I can determine whether a solution is reasonable.
- I can explain what a solution means in context.
- I can compare equation solving with other solution strategies.

What's To Come

This is a long-range preview for future lessons. Do not solve it today. Instead, annotate it individually. Circle anything familiar, underline symbols or directions that seem important, and write notes about what information you think will be needed later.

As you annotate, ask yourself: Which plan seems better right now? Could that change? What would you need to be sure?

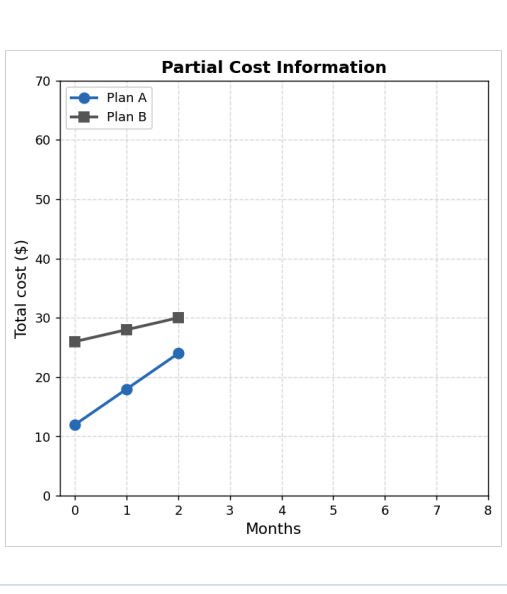
Future Task — Two-Plan Decision

A customer is comparing two phone protection plans. The table and graph below show only the first three months of data.

Partial Table (months 0–2 only)

Months	0	1	2
Plan A ($)	12	18	24
Plan B ($)	26	28	30

Partial Graph



Plan Descriptions

Plan A: Lower starting cost, higher monthly rate.

Plan B: Higher starting cost, lower monthly rate.

Decision Question

Which plan costs less over time? Can you make a confident recommendation right now?

1. Circle any numbers or patterns that seem useful for making a recommendation.
2. Underline one thing that makes it impossible to confidently recommend a plan today.
3. Describe what additional information or representation you would need. What specific question would it answer?

Intro

Directions

Read the introduction aloud with a partner or in a small group. As you read, annotate the text by highlighting important ideas, circling key vocabulary, and writing short notes about connections, questions, or predictions in the margins.

An equation is a mathematical sentence that says two quantities are equal. In a real situation, an equation can be used as a model. A model is not just a calculation. It is a way to represent what is happening.

When we write an equation from a situation, we need to decide what each variable means. We also need to decide which number is a starting amount, which number is repeated, and what total or target value matters.

Solving an equation gives a value. Interpreting the solution means explaining what that value represents in the situation. A number might be mathematically correct but still not make sense if the units or context do not fit.

In this section, we will connect equations, tables, graphs, and contexts so that solutions have meaning.

Stop and Discuss

1. The reading says an equation is a mathematical sentence. According to the reading, what does an equation say about the two quantities it contains?
2. The reading explains what interpreting a solution means. Find that sentence. What does the reading say could still be wrong even if the number is mathematically correct?
3. The reading says a model represents what is happening. Why might solving an equation correctly still give a useless answer in a real situation?

Equations Connect Situations to Solutions

MATH NOTES **Variables, Quantities, and Units**

Before writing or interpreting an equation, be specific about what each part represents.

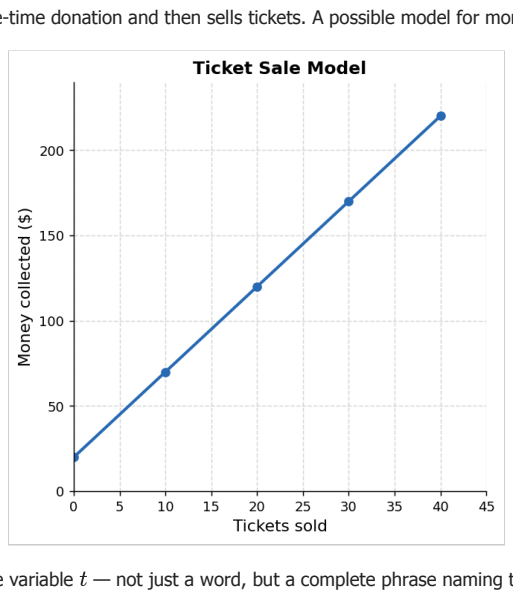
- A **variable** stands for a specific, changing quantity — not just a number.
- A **coefficient** is usually a rate: something that happens repeatedly for each unit of the variable.
- A **constant** is a fixed amount that happens once, regardless of the variable.
- A **solution** has units. "The answer is 5" is incomplete. "The answer is 5 months" or "5 tickets" is meaningful.

Strong variable definition: Instead of "x = tickets," write "x = the number of tickets sold."

YOUR NOTES

Example 1: Interpret the Model

The school drama club collects a one-time donation and then sells tickets. A possible model for money collected is $M = 5t + 20$.



1. Write a specific definition for the variable t — not just a word, but a complete phrase naming the quantity and its unit.
2. What part of the situation does the 5 represent? How does the graph support that interpretation?
3. What part of the situation does the 20 represent? What happens to M when $t = 0$ — and why does that match the context?
4. A different student writes $M = 20t + 5$. Explain specifically why that model does not fit the situation, using what 20 and 5 each actually represent.

You Try It 1: Define Before Writing

A bike rental shop charges a fixed fee when you pick up the bike, plus a cost for each hour rented.

1. Choose a variable for the number of hours. Write a specific definition — name the quantity and unit.
2. Choose a variable for the total cost. Write a specific definition.
3. List the two numbers you would need from the situation before you could write a complete equation. Explain which role each number plays (starting amount or repeated amount).
4. If someone gave you those two numbers, would you need anything else before writing the equation? Explain what would still be missing or assumed.

MATH NOTES **Reasonableness Before Solving**

Before writing and solving an equation, make a prediction about the answer.

- Should the answer be positive? What does a negative answer mean in this situation?
- Should the answer be a whole number? (People, tickets, and items usually are; time and cost may not be.)
- Is the target total larger or smaller than the starting amount? What does that tell you about whether the answer should be large or small?
- What unit should the answer carry?

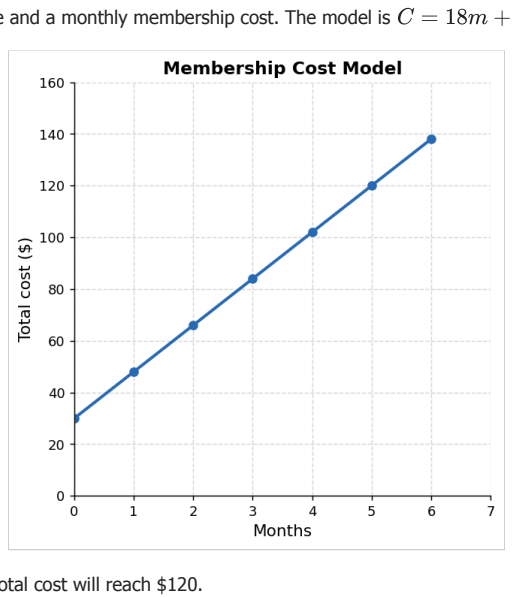
After solving, compare your result to your prediction. If they disagree, decide which one is wrong.

BEFORE SOLVING: MY PREDICTION

AFTER SOLVING: DOES IT MATCH?

Example 2: Think Before Solving

A fitness center charges a joining fee and a monthly membership cost. The model is $C = 18m + 30$.



A student wants to know when the total cost will reach $120.

1. Before solving, what does the variable m represent and what unit should the answer have? Use the graph to support your answer.
2. Is the target of $120 larger or smaller than the starting amount? Based on that comparison, estimate whether m will be closer to 1, 5, or 20 — and explain your reasoning without using algebra.
3. Would $m = 0.5$ be a reasonable solution for this situation? Explain why, even if the algebra produces that value.
4. Now write and solve the equation. Then interpret the solution in one complete sentence that includes the unit.

You Try It 2: Evaluate a Proposed Solution

A streaming service charges a setup fee and a monthly subscription. The model is $C = 9m + 15$.

> **Student claim:** "To find when the total cost reaches $60, I solved $9m + 15 = 60$ and got $m = 5$. So the answer is 5 dollars."

1. Check the arithmetic: does $m = 5$ satisfy the equation? Show the substitution.
2. Is the value $m = 5$ reasonable for this context? Would a value of $m = -2$ or $m = 0.5$ also be reasonable? Explain what each would mean.
3. The student's interpretation says "5 dollars." Write a corrected interpretation, and explain what specific error the student made when they named the units.

Stop and Discuss

Can two students get the same numerical solution but give different interpretations? Give a specific example — describe what the solution is and what each interpretation would say.

MATH NOTES **Critiquing Equation Models**

When evaluating a student's solution, errors can appear in three different places. Check each one separately.

1. **The model:** Does the equation correctly represent the situation? Do the numbers, variables, and operations match the context?
2. **The solving:** Were the algebraic steps correct? Do the solutions satisfy the equation when checked by substitution?
3. **The interpretation:** Does the answer make sense with units, context, and the question asked?

A student might get the correct number but misname the unit, or write a valid equation but mis-solve it. Identify exactly where the problem is.

WHAT TO CHECK

WHERE THE ERROR IS

Example 3: Critique the Interpretation

A community center charges $12 per class plus a $24 registration fee. A student writes $T = 12c + 24$ and solves $12c + 24 = 84$, getting $c = 5$.

> **Student conclusion:** "The solution means the total cost is 5 classes."

1. Use the context to verify that the equation $T = 12c + 24$ correctly models the situation. Identify what each part represents.
2. Verify that $c = 5$ satisfies the equation by substituting. Does the equation error, solving error, or interpretation error apply here?
3. What is specifically wrong with the student's conclusion? The error is not in the number — what is wrong?
4. Write a corrected conclusion in a complete sentence that includes units and explains what the solution actually means.

You Try It 3: Revise the Model

A school club orders shirts. There is a one-time $40 design fee and each shirt costs $8.

> **Student work:** "I wrote $8(s + 40) = 200$. I solved and got $s = -15$, so the club should order negative 15 shirts."

1. Use the three-part critique framework. Where is the error — model, solving, or interpretation?
2. Distribute $8(s + 40)$ and compare the result to what the context says. What specifically went wrong when the student set up the equation?
3. Write a corrected equation for the situation. Solve it and explain how the corrected model gives a different and more meaningful solution.
4. The student's interpretation was also wrong. Write what a corrected interpretation would say for the actual solution.

MATH NOTES **Building and Comparing Models**

To build an equation model from a situation:

1. Name and define each variable.
2. Identify the starting amount (constant — happens once).
3. Identify the rate (coefficient — happens per unit of the variable).
4. Write the equation.
5. Use the equation to answer the question, then interpret with units.

To compare two models, ask: Which model gives a lower cost at the beginning? Which model gives a lower cost after many time periods? Is there a point where they are equal?

YOUR NOTES

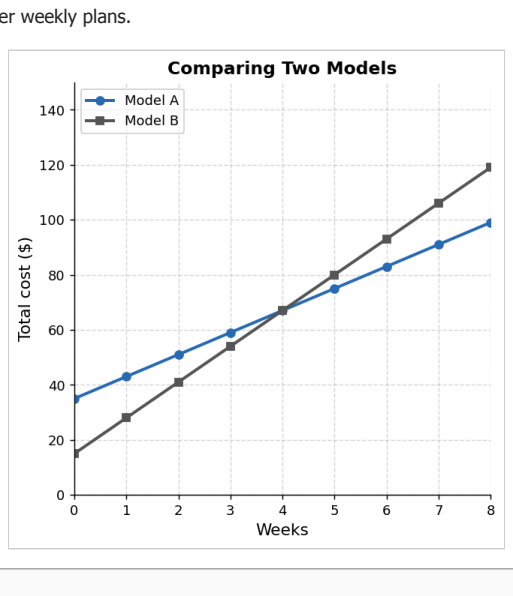
Example 4: Build the Model

A school fair charges a $10 entry fee and $3 for each ride ticket. Let r represent the number of ride tickets purchased and C represent the total cost.

1. Write an equation for C in terms of r. Identify which number is the starting amount and which is the rate.
2. Use your equation to find the total cost for 12 ride tickets. Show the substitution.
3. Write a new equation to find how many ride tickets a person bought if their total cost was $49. Explain what is different about this equation compared to the one in Q1.
4. Solve your equation from Q3 and interpret the solution with units. Is the solution reasonable?

You Try It 4: Compare and Defend

Two equipment rental companies offer weekly plans.



Company A: $C = 8w + 35$

Company B: $C = 13w + 15$

1. In each equation, what does the constant term represent in the rental context? Is the constant a rate or a fixed amount? Explain how you know.
2. At $w = 3$ weeks, which company costs less? Show the calculation for both and compare.
3. The graph shows the two lines cross somewhere between weeks 3 and 4. Without finding the exact crossing point, explain what happens to the relative cost of Company A vs. Company B before and after the crossing. Which equation supports a longer rental — and why?
4. A customer plans to rent for 6 months. Write a recommendation letter sentence that names which company they should choose, gives the specific cost at 6 months for each company, and explains why.

Summary

An equation model is strongest when the equation, variable definitions, solving steps, units, and interpretation all refer to the same situation. Writing an equation without defining variables, or solving without interpreting, leaves the work incomplete.

When evaluating a solution, check three places separately: the model, the solving, and the interpretation. An error in any one of these places makes the conclusion unreliable, even if the other two are correct.

When comparing models, ask which is better for the specific question. A lower starting cost does not mean lower total cost — the rate matters more over time.

Common Mistakes

- **Solving correctly but naming the wrong unit.** Why it is tempting: the algebra produces a number, so the work feels finished. Check instead: ask what the variable represents and write the answer in a complete sentence with units.
- **Writing an equation before defining variables.** Why it is tempting: it feels faster to use x immediately. Check instead: name the quantity and unit for each variable before writing the equation.
- **Accepting an unreasonable solution.** Why it is tempting: negative or non-integer answers can appear during algebra. Check instead: decide whether the answer makes sense for people, tickets, months, or items — and if not, look for a model error.
- **Confusing starting amount and rate.** Why it is tempting: both numbers appear in the context. Check instead: ask which amount happens once (constant) and which happens for each unit (coefficient).
- **Assuming a lower constant term means a better model.** Why it is tempting: a smaller number up front looks cheaper. Check instead: compare total cost at the specific time period in the question — the rate can make one model more expensive over time.

Optional Future Task Reflection

Look back at the future task. Do not solve it yet. Highlight one part that makes more sense now than it did at the beginning. Circle one part that still feels unfamiliar. Write one sentence about what representation you would look at first.